Ex 1. (12/05/2026)

Consider a 1-period, 2-state model like

$$S \xrightarrow{p} S_u, \quad S \xrightarrow{1-p} S_d$$

where $S = 100$, $S_u = 150$, $S_d = 75$.

Assume that you know the real-world probability $p = \frac{1}{2}$. Assume $r = 0$ interest rate.

Consider a European call option C with $K = 110$.

(a) Show that pricing this option with the risk-neutral valuation principle gives different results than with real-world probabilities

(i) Price with real-world probabilities:

Denote C^P for the price of C under $\mathbb{P}$.

$$C^P = \frac{1}{1+r}\left(p\,C_u + (1-p)\,C_d\right)$$

with $C_u = (S_u - K)_+ = 40$, $C_d = (S_d - K)_+ = 0$

So,

$$C^P = 1 \cdot \left(\tfrac{1}{2} \cdot 40 + \tfrac{1}{2} \cdot 0\right) = 20.$$

(ii) Price with risk-neutral probabilities:

Denote C^q for the price of C under $\mathbb{Q}$.

$$C^q = \frac{1}{1+r}\left(q\,C_u + (1-p)\,C_d\right)$$

with

$$q = \frac{S(1+r) - S_d}{S_u - S_d} = \frac{100\,(1+0) - 75}{150 - 75} = \frac{25}{75} = 0.33$$

Thus

$$C^q = 1 \cdot \left(\tfrac{1}{3} \cdot 40 + \tfrac{2}{3} \cdot 0\right) = \frac{40}{33} = 13.33$$

Hence $13.33 = C^q < C^P = 20$

$\Rightarrow$ only risk-neutral measure should be used for option pricing since it does not introduce arbitrage.

(b) Show how you could construct an arbitrage strategy if someone would transact the call option at the price, using the real world probabilities.

From theorem 1: $C^q = H^* = \alpha^* + \Delta^* S$

with

$$\Delta^* = \frac{C_u - C_d}{S_u - S_d} = \frac{40-0}{150-75} = \frac{8}{15} \approx 0.53$$

$$\alpha^* = \frac{1}{1+r}\left(C_u - \frac{C_u - C_d}{S_u - S_d} S_u\right) = 1 \cdot \left(40 - \frac{40-0}{150-75} \cdot 150\right) = -40$$

Hence $H^* = -40 + 0.53 \cdot 100 = 13.33 = C^q$.

* The arbitrage strategy:

By the assumption, there exists a person that wants to buy the call option at price C^P (since he believes that this is the correct price).

Consider the following strategy:

Time $t=0$:

- Sell the call to this person at premium C^P: $+C^P = +20$
- Buy H^*, so borrow €40 and buy 0.53 shares of S:
  $-H^* = -C^q = -13.33$
- Invest the difference in risk-free asset at $r=0$: -6.67

$\Rightarrow$ initial wealth is €0.

Time $t=1$:

(i) If $S_1 = S_u$:

- Person exercises the call: $-(S_u - K)_+ = -40$
- We sell H^*, it is worth $\alpha^*(1+r) + \Delta^* S_u = -40 + 0.53$
  $= 40$
- We receive risk-free asset investment: $+6.67$

$\Rightarrow$ Final wealth is $-40 + 40 + 6.67$
$= €6.67$

(ii) If $S_u = S_d$:

- Person does not exercise the call: -0
- We sell H^*, it is worth $\alpha^*(1+r) + \Delta^* S_d = -40 + 0.53 \cdot 75$
  $= 0$
- We receive risk-free asset investment: $+6.67$

$\Rightarrow$ Final wealth is $-0 + 0 + 6.67$
$= €6.67.$

Conclusion:

If someone is willing to buy the call at $C^P = €20$, then we can construct a portfolio with initial cash 0 and final cash $+6.67$

$\Rightarrow$ Arbitrage!